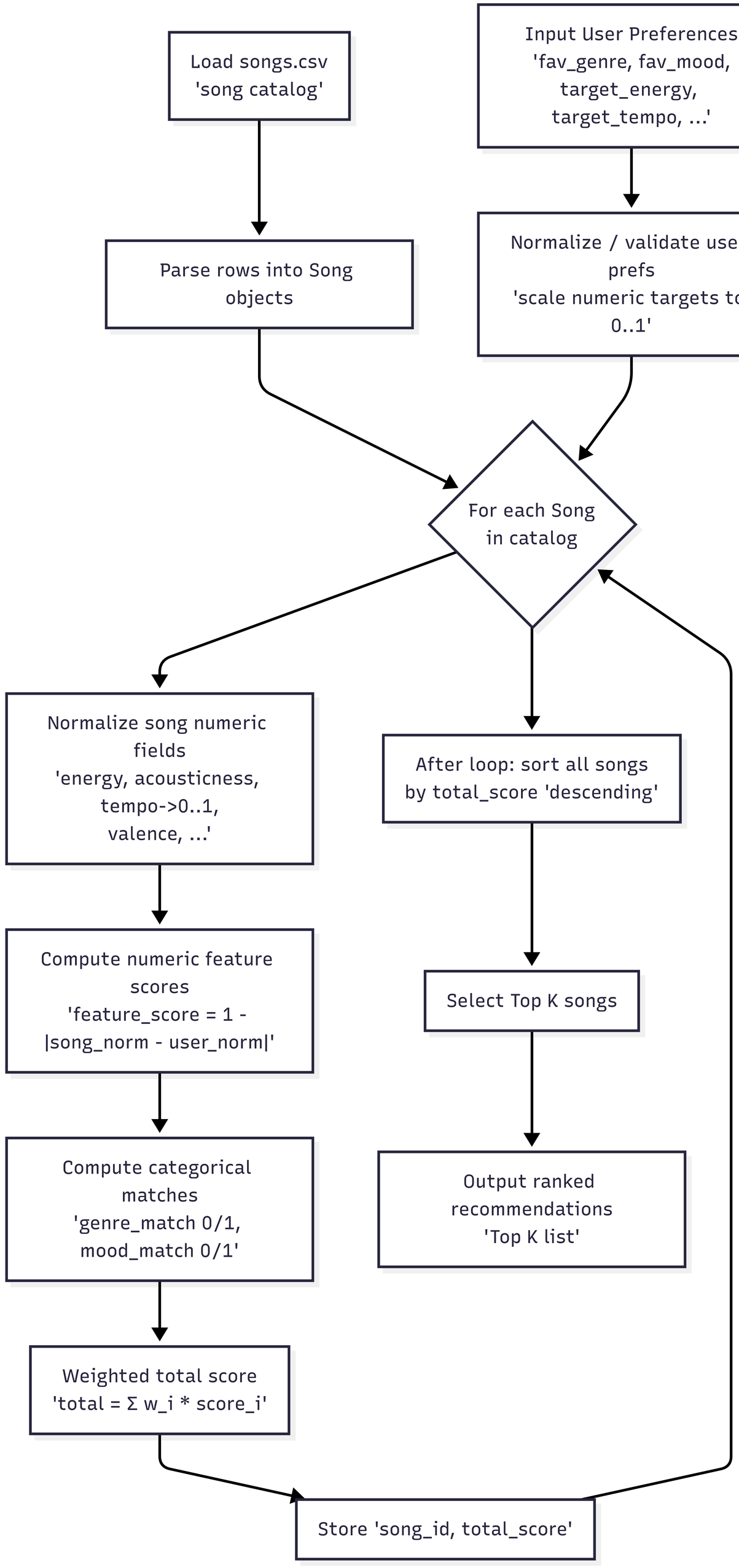
Load songs.csv 'song catalog'
Input User Preferences 'fav_genre, fav_mood, target_energy, target_tempo, ...'
Parse rows into Song objects
Normalize / validate user prefs 'scale numeric targets to 0..1'
For each Song in catalog
Normalize song numeric fields 'energy, acousticness, tempo->0..1, valence, ...'
After loop: sort all songs by total_score 'descending'
Compute numeric feature scores 'feature_score = 1 - |song_norm - user_norm|'
Select Top K songs
Compute categorical matches 'genre_match 0/1, mood_match 0/1'
Output ranked recommendations 'Top K list'
Weighted total score 'total = Σ w_i * score_i'
Store 'song_id, total_score'